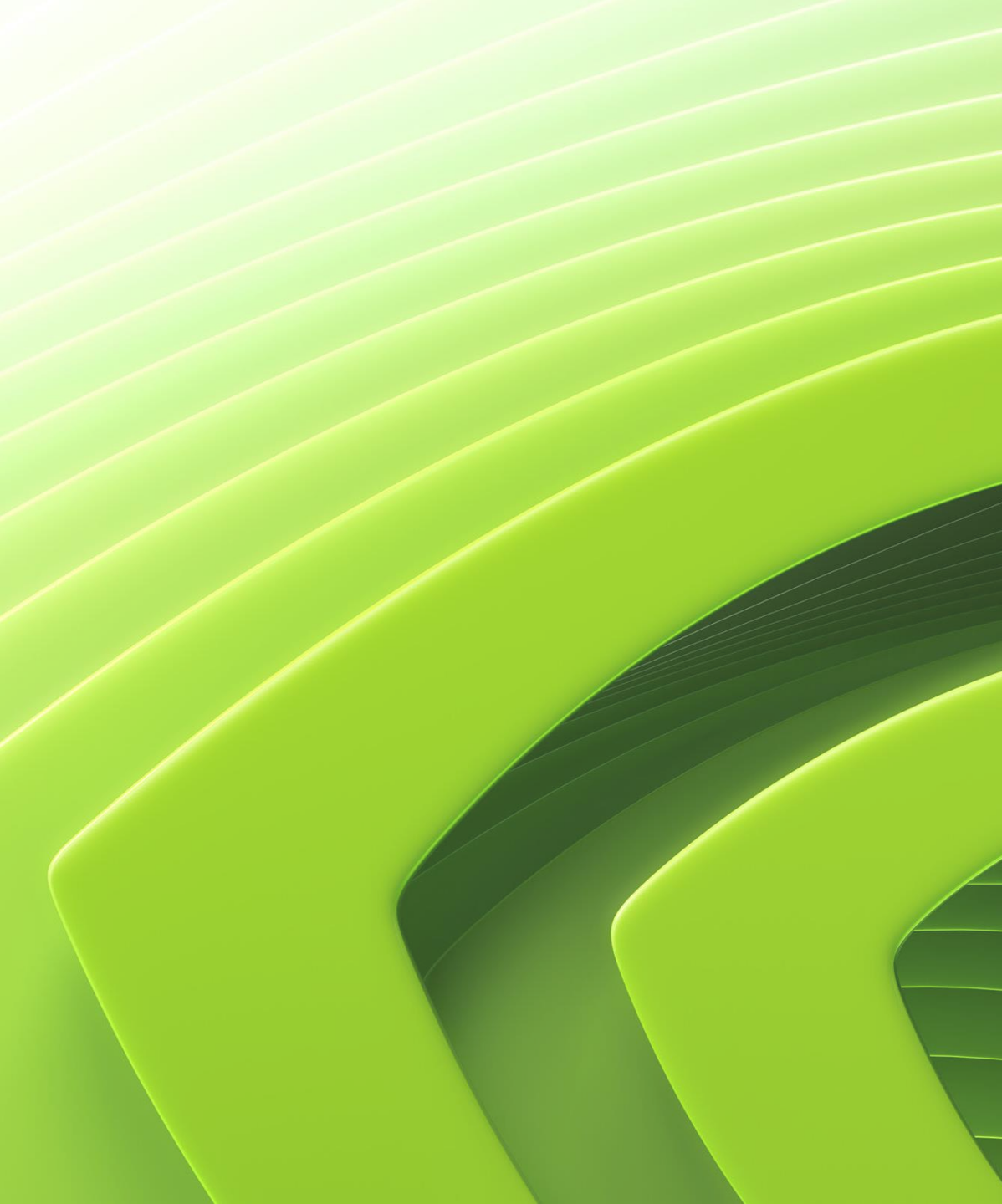




Accelerate Robotics and Real-Time AI Inference on NVIDIA Jetson Thor

Oct 15 2025



AGENDA

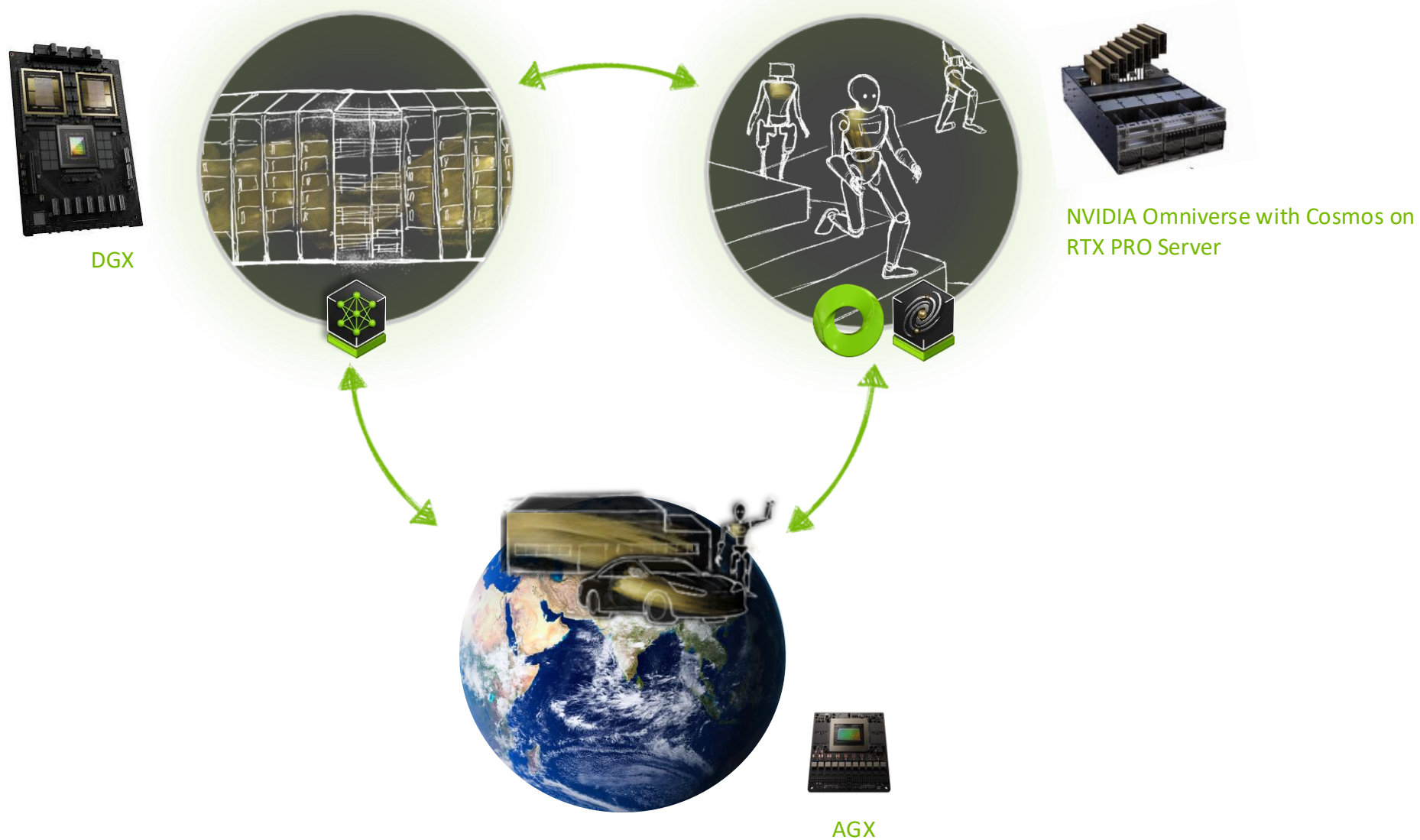
Introduction to Jetson Thor Platform and its unique innovations

Accelerate development with Jetson Thor Software Stack

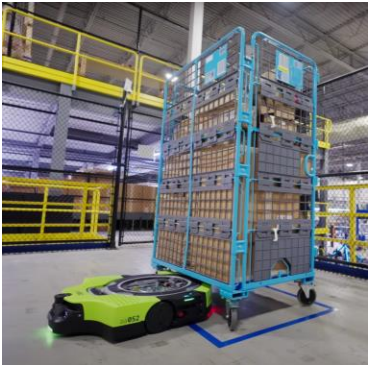
Boost GenAI performance 7X with the new vLLM container and speculative decoding

Faster application creation with NVIDIA Isaac, Metropolis VSS and more.

NVIDIA's 3 Computers to Build Physical AI



Generative AI Revolutionize Robotic Industry



Robots Today
Single Purpose



Pick up socks from the grey tote and place in the empty box on the conveyor

Pack the boxes in the blue cart so there is no empty space left

Take this cart to station number 9

Pickup box number 11 from the shelf and place it on the conveyor

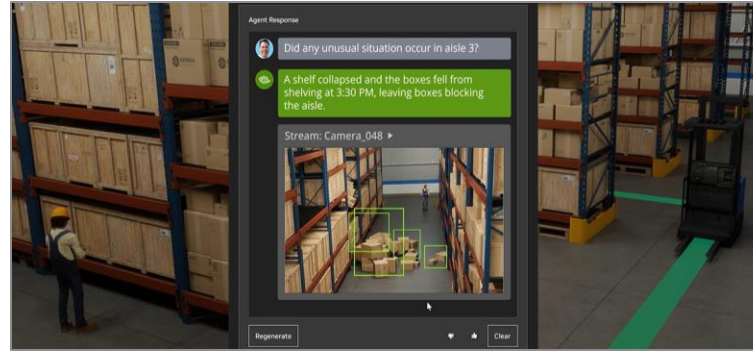
Pick one package at a time from conveyor and place it on the sorting station

Robots Tomorrow
General Purpose

Real-time Reasoning is the Key to Unlock Physical AI and Robotics



Humanoid Robotics
Multimodal AI



Visual Agent
Custom Alert and Summarization



Medical Robotics and Instrument
Realtime Sensor Processing



Autonomous Robot and Machine
Localization and Navigation



Edge HPC and Inference
High-Bandwidth Signal Processing

Introducing NVIDIA Jetson Thor

The ultimate platform for physical AI and robotics

- 2070 FP4 | 1035 FP8 TFLOPs powered by Blackwell GPU
 - New Transformer engine architecture
 - Multi-Instance GPU (MIG) support
- 14-core Neoverse ARM CPU for real-time processing
- 128 GB 256-bit LPDDR5X memory, 273 GB/s for large transformer models
- 4X 25GbE for high-speed sensor fusion
- 40W-130W for entire module
- Jetson Thor Developer Kit Priced at \$3499
- *Available Now!*



Introducing NVIDIA Jetson Thor

The ultimate platform for physical AI and robotics



7.5x Orin
AI Compute



3.5x Orin
Energy Efficiency



3.1x Orin
CPU Performance



10x Orin
I/O Throughput



Jetson Thor Module Specifications

	JETSON AGX ORIN 64GB	JETSON T5000	JETSON T4000
GPU ARCH	AMPERE	BLACKWELL	
GPU CORES	2 GPC, 2048 CUDA Cores 64 Tensor Cores	3 GPC, 2560 CUDA Cores 96 Tensor Cores MIG Support	2 GPC, 1536 CUDA Cores 64 Tensor Cores MIG Support
GPU Max Frequency	1.3 GHz	1.6 GHz @ 85C	
PEAK AI PERF	275 INT8 TOPS (Sparse) 138 INT8 TOPS (Dense)	Up to 2070 FP4 SPARSE TFLOPs Up to 1035 FP4 DENSE TFLOPS	Up to 1200 FP4 SPARSE TFLOPs Up to 600 FP4 DENSE TFLOPS
CPU	12X A78 @ 2.2 GHz	14x Neoverse @ 2.6 GHz	12x Neoverse @ 2.6 GHz
CODEC	1X NVDECODE / 1X NVENCODE	2X NVDECODE / 2X NVENCODE	1X NVDECODE /1X NVENCODE
VISION ACC	PVA v2	PVA v3	
DRAM SIZE	64GB	128 GB	64 GB
DRAM BW	3200 MHz 256-bit → 204 GB/s	4266 MHz 256-bit→273 GB/s	
Networking	10 GbE	4x 25GbE	3x 25GbE
Display	1x 8K60 multi-mode DP 1.4a (+MST)/eDP 1.4a/HDMI 2.1	4x shared HDMI2.1/ VESA DisplayPort 1.4a—HBR2, MST	
Camera	Up to 6 cameras (16 via virtual channels) 16 lanes MIPI CSI-2	Up to 20 cameras via HSB Up to 6 cameras through 16x lanes MIPI CSI-2 Up to 32 cameras using Virtual Channels	
PCIe	Up to 2 x8 + 1 x4 + 2 x1 (PCIe Gen4, Root Port, & Endpoint)	Up to Gen 5 x8 lanes Root Port Only – C1 (x1) and C3 (x2) / Root Point or Endpoint – C2 (x1), C4 (x8), and C5 (x4)	
USB	3x USB 3.2 Gen2 (10 Gbps) 4x USB 2.0	xHCI host controller with integrated PHY (up to) 3x USB 3.2 4x USB 2.0	
Other IO	4x UART, 3x SPI, 4x I2S, 8x I2C, 2x CAN, PWM, DMIC & DSPK, GPIOs	5x I2S/2x audio hub (AHUB), 2x DMIS, 4x UART, 4x CAN, 3x SPI, 13x I2C, 6x PWM outputs	5x I2S/2x audio hub (AHUB), 2x DMIS, 4x UART, 3x SPI, 13x I2C, 6x PWM outputs
MECHANICAL	100x87 mm , 699 pins	100x87 mm , 699 pins	
MODULE MAX POWER	60W	130W	70W

NVIDIA Jetson Thor

Hardware Collaterals

The screenshot shows the NVIDIA Developer website's Jetson Download Center. The header includes the NVIDIA Developer logo and navigation links: Home, Blog, Forums, Docs, Downloads, and Training. A secondary navigation bar lists categories: Jetson, Developer Kits, Production Modules, Software, Partners, Community, and Buy. The main heading is "Jetson Download Center". Below it, a message says "See below for downloadable documentation, software, and other resources." A section titled "New to the Jetson platform?" provides links to FAQ, support resources, tutorials, and online documentation, along with a list of documents to start with: Jetson Developer Kit user guides, Jetson module datasheets, Jetson Linux Developer Guide, Pin and function names guides, and Jetson PCN Center. A search bar on the left contains the text "Thor" and has a "Clear All Filters" button. To the right, a table displays search results for "Thor", showing 18 of 307 downloads. The table has columns for Title, Version, and Release Date. The results list various documents such as the Jetson AGX Thor Developer Kit Carrier Board Specification, Jetson T5000 and Jetson AGX Orin Series Interface Comparison and Migration Application Note, Jetson T5000 Series Modules Data Sheet, Jetson Thor Series Modules Pinmux Template, Jetson Thor Series Modules Fuse Specification, Jetson Thor Series Developer Kit 3D CAD STEP model, and Jetson Thor Series Modules Regulatory and Compliance Documents.

Find older items
[Jetson Downl](#)
[Archive](#)

Jetson Download Center

See below for downloadable documentation, software, and other resources.

New to the Jetson platform?

- Please read the [FAQ](#), check out our [support resources](#), [tutorials](#), and browse the [online documentation](#)
- Documents to start with are:
 - [Jetson Developer Kit user guides](#)
 - [Jetson module datasheets](#)
 - [Jetson Linux Developer Guide](#)
 - [Pin and function names guides](#)
 - [Jetson PCN Center](#)

Showing 18 of 307 downloads. Filtering by "Thor".

Title	Version	Release Date
> Jetson AGX Thor Developer Kit Carrier Board Specification	1.1	2025/09/17
> Jetson T5000 and Jetson AGX Orin Series Interface Comparison and Migration Application Note	1.2	2025/09/17
> Jetson T5000 Series Modules Data Sheet	1.2	2025/09/17
> Jetson Thor Series Modules Pinmux Template	1.3	2025/08/22
> Jetson Thor Series Modules Fuse Specification	1.0	2025/08/21
> Jetson Thor Series Developer Kit 3D CAD STEP model	V01	2025/08/12
> Jetson Thor Series Modules Regulatory and Compliance Documents	1.0	2025/08/05

Jetson Download Center - Thor Series

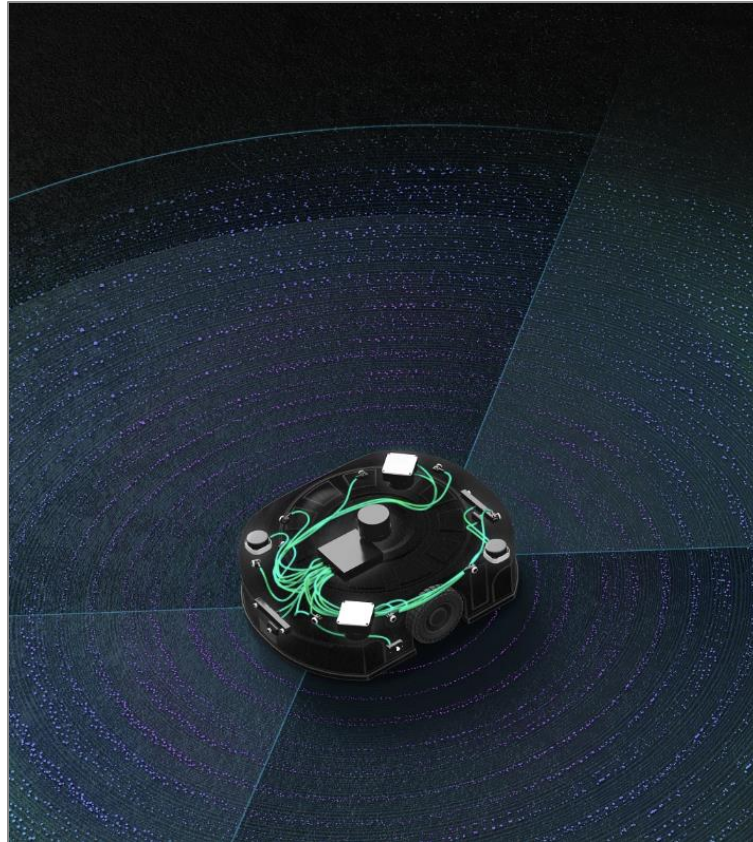
- [Jetson T5000 Datasheet](#)
- [Jetson Thor Series Modules - Design Guide](#)
- [Jetson Thor Series Modules - Thermal Design Guide](#)
- [Jetson Thor Series Modules – Tuning Guide](#)
- [Jetson AGX Thor Developer Kit Carrier Board Specification](#)
- [Jetson Thor Series Pinmux Template](#)
- [Jetson Thor Series Modules Fuse Specification](#)
- [Jetson Thor Series Supported Components List](#)
- [Jetson Thor Series Product Marking Specification](#)

Challenges in Real-time Sensor Processing



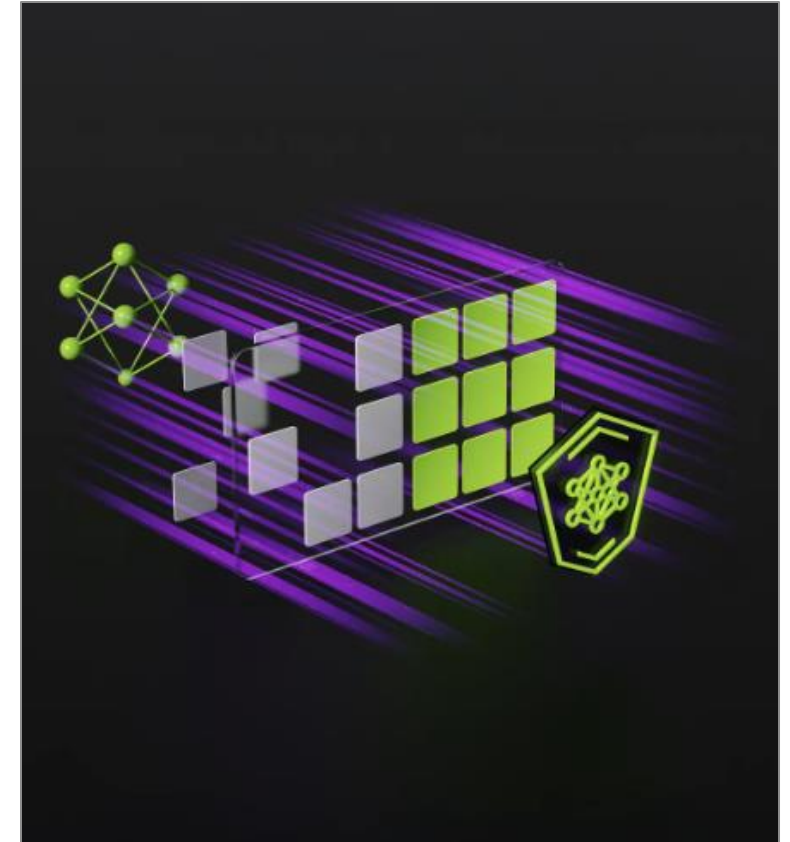
Latency Measured in Microseconds

Modern Industrial and Edge Solution Demands Ultra Low Latency from Photon to Memory



Time Consuming Sensor Integration

Software-Defined Sensors are required to make easy for driver development and integration

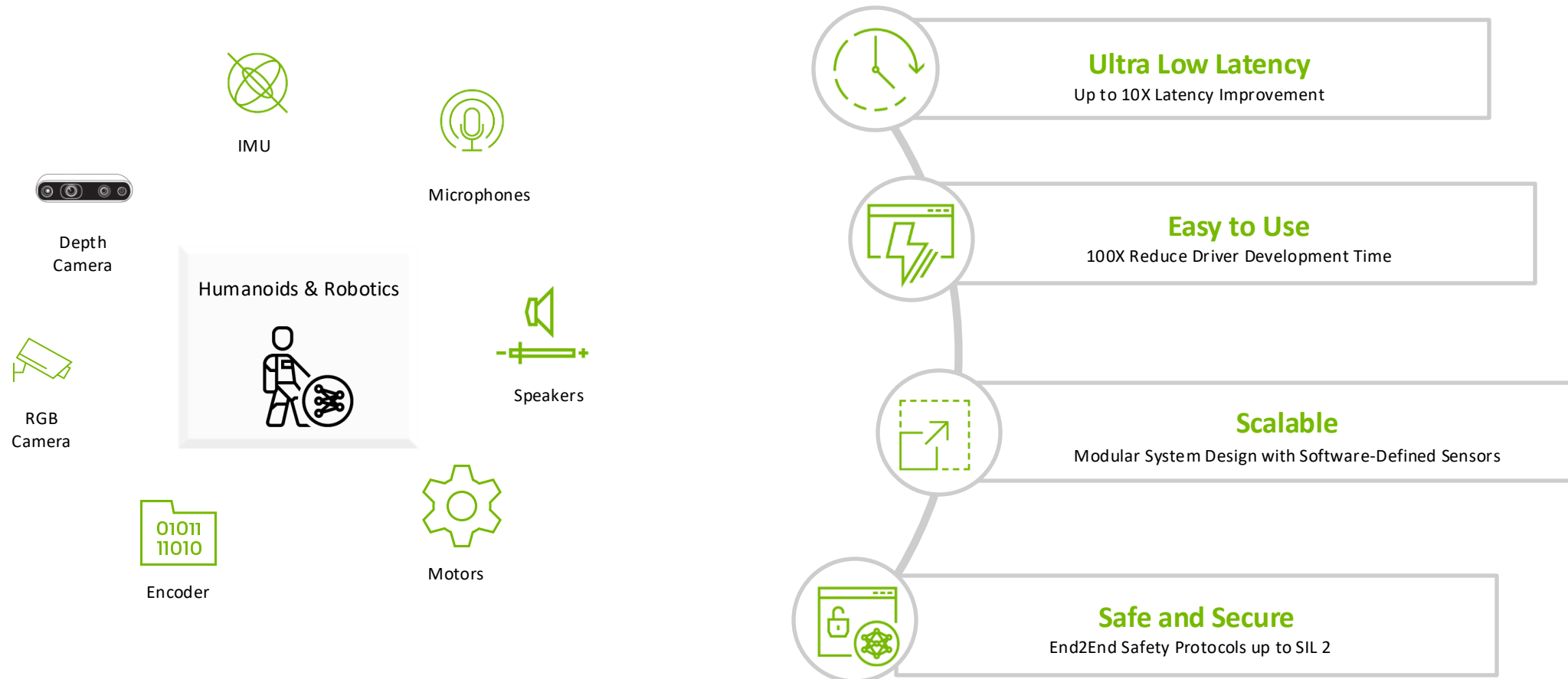


Security and AI Safety

Increased data flow necessitates advanced cybersecurity measures and AI safety guardrail

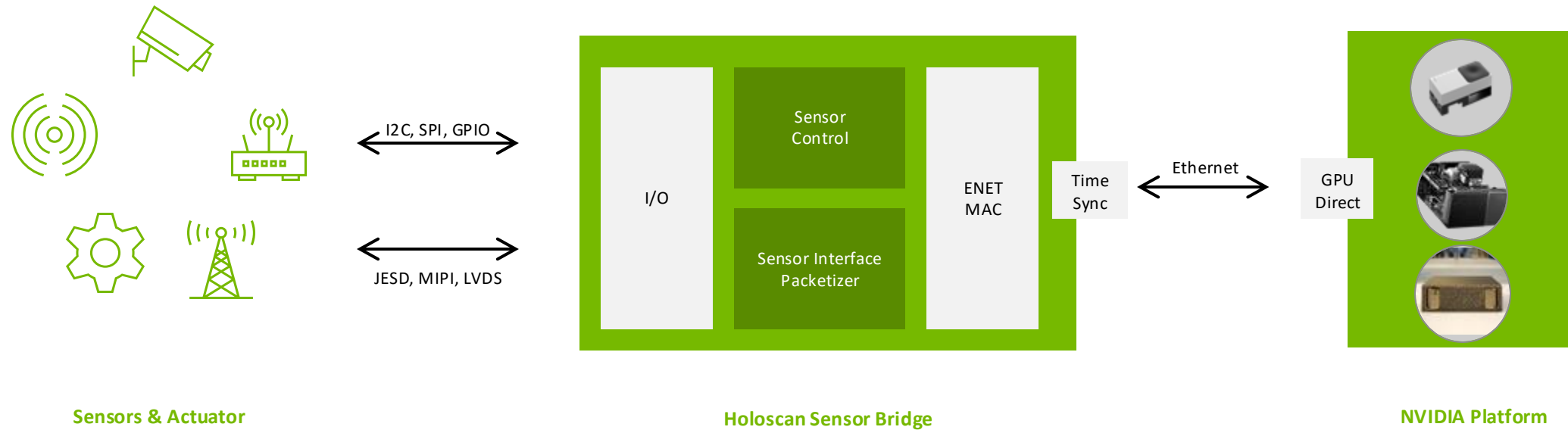
Introducing Holoscan Sensor Bridge

Sensor-to-Ethernet Streaming Platform



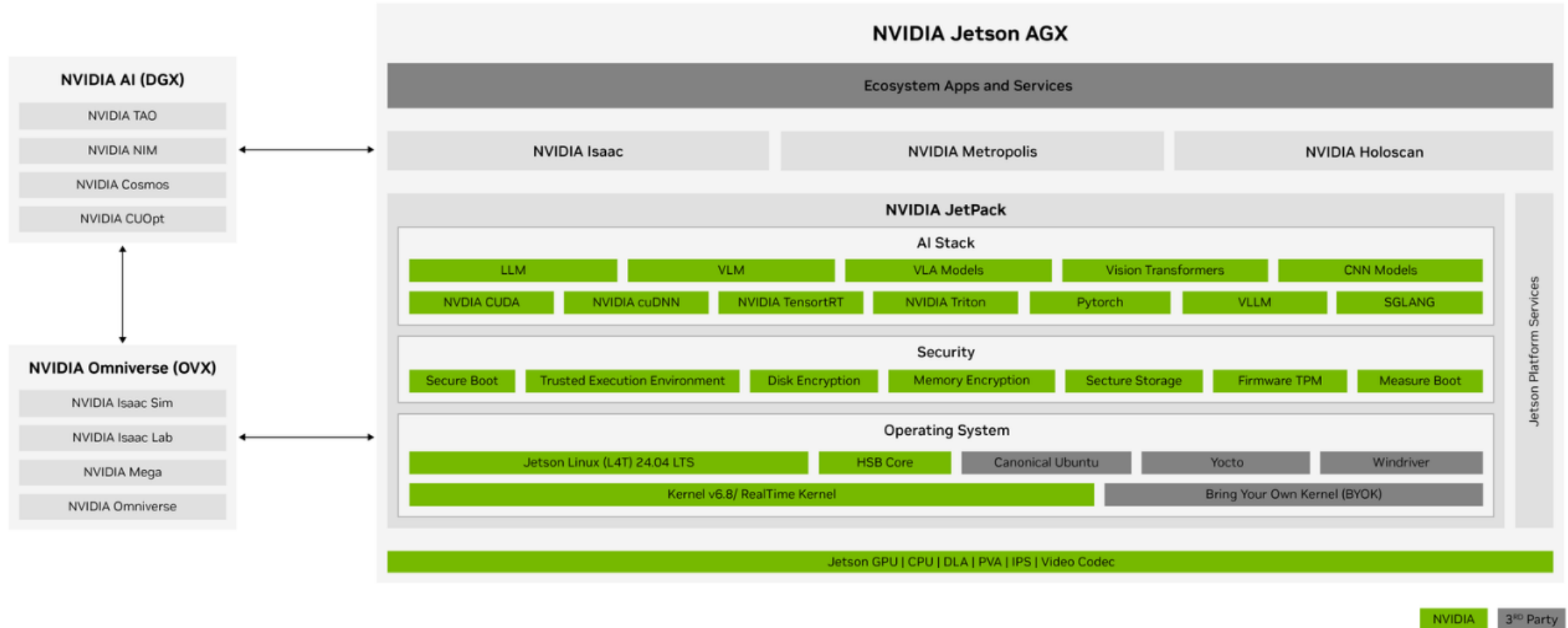
Learn More: <https://www.nvidia.com/en-us/technologies/holoscan-sensor-bridge/>

Holoscan Sensor Bridge Reference Design Architecture



NVIDIA Jetson Thor Software Stack

Optimized for Humanoid, Robotics and Generative AI Reasoning



JetPack 7 is Production-Ready

High Level	Distro/Kernel	Security	Camera
Blackwell architecture support Multi-Instance GPU New SBSA Based Architecture New Integrated HSB New Modular Architecture ISO image-based installation New	Ubuntu 24.04 LTS New Kernel 6.8 LTS New Realtime Kernel New	Secure Boot Trusted Execution Environment Disk Encryption Memory Encryption Firmware TPM Measured Boot FSKP Support New	HSB Camera New MIPI Camera Ethernet Camera IMX Camera USB Camera Argus V4L2 SIPL New Jetson IQ tuner utility
Multimedia	Display	Graphics	Power Management
2-NVENC 2-NVDEC 2-NVJPG 1-OFA Unified Video Driver (IGPU-DGPU) NVIDIA MMAPAPI Buffer Utility GStreamer NVOsd Onscreen Display	Up to 4xHDMI Up to 4xDP DP/HDMI Mix and Match New Direct Rendering Manager	GLX EGL X11 OpenGL-ES Vulkan Weston Open GL Wayland EGLDevice OpenWFD	CPU/GPU Low power states Dynamic frequency scaling IO Runtime Suspend Suspend to RAM (SC7) Passive and Active cooling Power Capping NVP Models Power Estimator Tool

NOTE:
1. This is not an exhaustive list. Please visit Jetson Linux Developer guide for documentation
2. Some features will be delivered with JetPack 7.x releases

Installation & Flashing

New ISO Based Installation Added

- **Developer Kits**

- ISO Based Installation (No need for HOST PC!)
 - ISO Installation installs OS only. For compute stack, do “apt install nvidia-jetpack”
 - Refer [here](#) for Quick Start Guide.
- SDK Manager
- Flash script

Note: If someone is re-installing BSP using ISO, there is an extra step. Please refer [here](#).

- **Modules**

- Manual Flashing. Only support Initrd Flashing. Refer [here](#).
 - Note there is a difference for Thor:
 - `sudo ./apply_binaries.sh --openrm`
 - Flash command: `sudo ./l4t_initrd_flash.sh jetson-agx-thor-devkit internal`

Table of Contents

Getting Started

Quick Start Guide

Detailed Setup Guides

BSP Installation

Docker Setup

CUDA Setup

JetPack SDK Setup

Hardware

Hardware Layout

Supported Hardware

Extra Guides

Interim Solutions

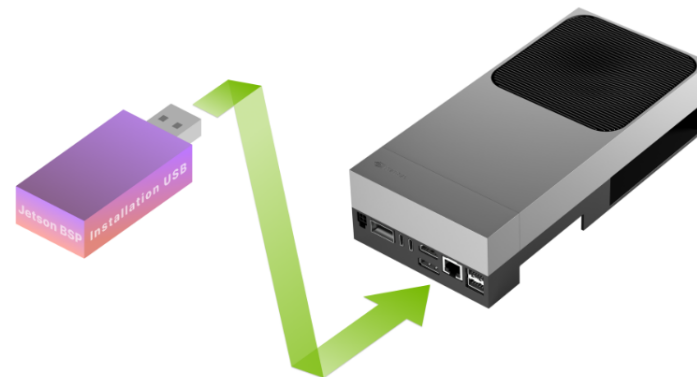
Troubleshooting Guide

Resources

Additional Docs

Quick Start Guide

Quick Start Guide



On this page

Overview

Steps

- 1 Download ISO image
- 2 Create installation USB
- 3 Unboxing and setup
- 4 Boot from the USB stick and install the BSP on NVMe
- 5 Boot BSP from NVMe for the first time

Your Next Step

- ⚠ Reinstall with USB (Previously Used Device)
- 🔧 BSP Install Troubleshoot

Overview

In this guide, we will walk you through the steps to quickly install the BSP (*Jetson Linux*) on your Jetson AGX Thor Developer Kit using a **bootable installation USB stick**.

It's a new process introduced with JetPack 7.0 that enables you to install the Jetson BSP (*Jetson Linux*) **without using a host Ubuntu PC**, making the initial setup process simpler and similar to how you would install Ubuntu on your laptop or a PC.

Attention

Jetson's bootable installation USB stick employs a very minimal version of Linux and made purely for the purpose of installing the Jetson BSP (*Jetson Linux*) on Jetson AGX Thor Developer Kit.

Unlike Canonical's official Ubuntu installation USB, it does NOT offer the capability of booting into full Jetson BSP for the purpose of test drive, thus **it is not "Live USB" stick** in that sense.

What you'll need

- A laptop or PC (Windows, Mac, or Linux) with at least 25GB of storage space
- A USB flash drive (16GB or larger)

Overall Flow

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Getting Started

[Quick Start Guide](#)

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[Troubleshooting Guide](#)

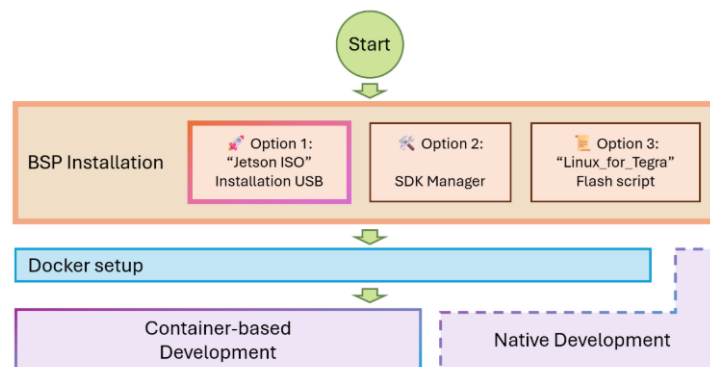
Resources

[Additional Docs](#)

[Home](#) > [BSP Installation](#)

BSP Installation

BSP Installation Options



Comparison

	 Jetson ISO	 SDK Manager	 Flash script
 Short Summary	USB installation disk to flash BSP (plus alpha)	GUI software to flash BSP and install JetPack packages	Ubuntu script to flash BSP
 Ubuntu Host PC	 Not required	 Required	 Required
 Flash time	 Less than 15 min	 About 30 min	 About 30 min
 Install BSP?	 Yes	 Yes	 Yes
 Install Docker?	 Yes	 Yes, when chosen in 2nd stage	 No
 Install JetPack components	 No	 Yes, when chosen in 2nd stage	 No
 Who is this for?	 Everybody including beginner	 Who has an Ubuntu PC	 Product Developers

Option 1. "Jetson ISO" Installation USB

On this page

BSP Installation Options

[Comparison](#)

Option 1.  "Jetson ISO" Installation USB

Option 2.  NVIDIA SDK Manager

Option 3.  "Linux_for_Tegra" flash script

SBSA and OpenRM

- Jetson Thor uses Server Based System Architecture (SBSA)
 - Moved to OpenRM from nvgpu driver
 - Same installation bits and toolchains as desktop and server platforms.
 - Same containers can be used, for example, on Spark and GB200.
 - Streamlines developer experience across NVIDIA platforms.
 - Unified CUDA toolkit between desktop GPU and Jetson embedded.

CUDA 13 Features

- Available in JetPack 7.0 (Blog: <https://developer.nvidia.com/blog/whats-new-in-cuda-toolkit-13-0-for-jetson-thor-unified-arm-ecosystem-and-more/>)
 - SBSA for Thor
 - Unified Virtual Memory with Full Coherence
 - Multi-Process Service (MPS)
 - Deterministic GPU scheduling with Green Contexts
 - Simplified memory sharing with DMA-BUF
 - Support for Non-Uniform Memory Access (NUMA) aware apps
- JetPack 7.1
 - Minor release for bug fix and support for Jetson T4000 module
- JetPack 7.2
 - Multi Instance GPU
 - MIG_0 : 4 (GPC_0), MIG_1 : 6 (GPC_1 + GPC_2)
 - Graphics is only on MIG_0

Camera API's

Argus

- Argus is the main Api to work with MIPI/CSI cameras.

SIPL

- SIPL is the main Api for Holoscan Sensor Bridge (HSB) cameras.

Feature/Aspect	SIPL	Argus
Full Name	Safe Image Processing Library	NVIDIA Argus Interface with HW ISP functionality Providing per frame control.
Purpose	Unified API and driver framework for diverse camera hardware, especially Camera-over-Ethernet (CoE), i.e, HSB Cameras	Interface to CSI cameras with Tegra Multimedia API focusing on ISP functionalities
Supported Camera Types	Primarily Camera-over-Ethernet (CoE), supports Holoscan Sensor Bridge (HSB) based sensors	CSI cameras connected via MIPI CSI interface [Also supports GMSL]
Architecture/Design	Modular, extensible, JSON configuration driven, uses SIPL API as backend.	C++ interface, uses <u>LibArgus</u> and Tegra MM API as backend.
Camera Drivers	User mode camera sensor drivers.	Kernel mode camera sensor drivers.
Hardware Support	<u>CoE</u> sensors, other transport abstractions (MIPI, GMSL, etc for future)	CSI sensors, GMSL.

Jetson Thor Supports All Modern Gen AI Frameworks and Models

AI Frameworks



Hugging Face



LLaMA



MACHINE LEARNING
COMPILATION



NVIDIA

TensorRT-LLM



ONNX

PyTorch

SGL



vLLM

Generative AI Models



ByteDance



deepseek

Gemini

Meta



MISTRAL
AI



OpenAI

Physical Intelligence (π)

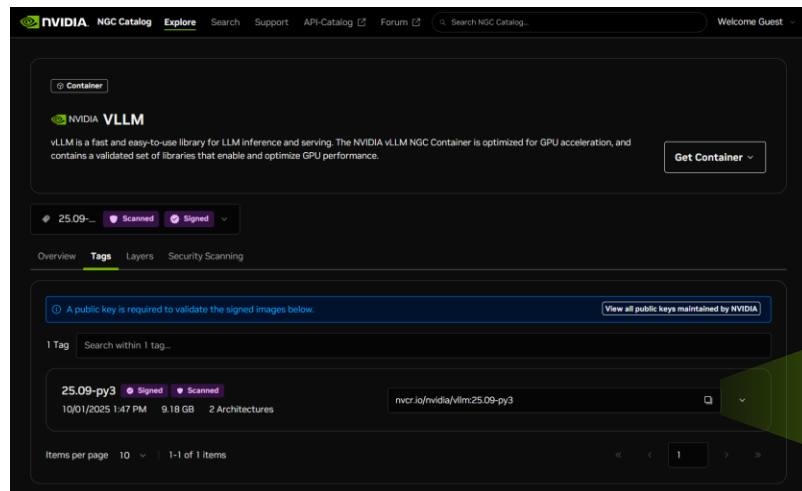


Qwen

LLM Inference Serving Options

Officially Supported and Community Supported

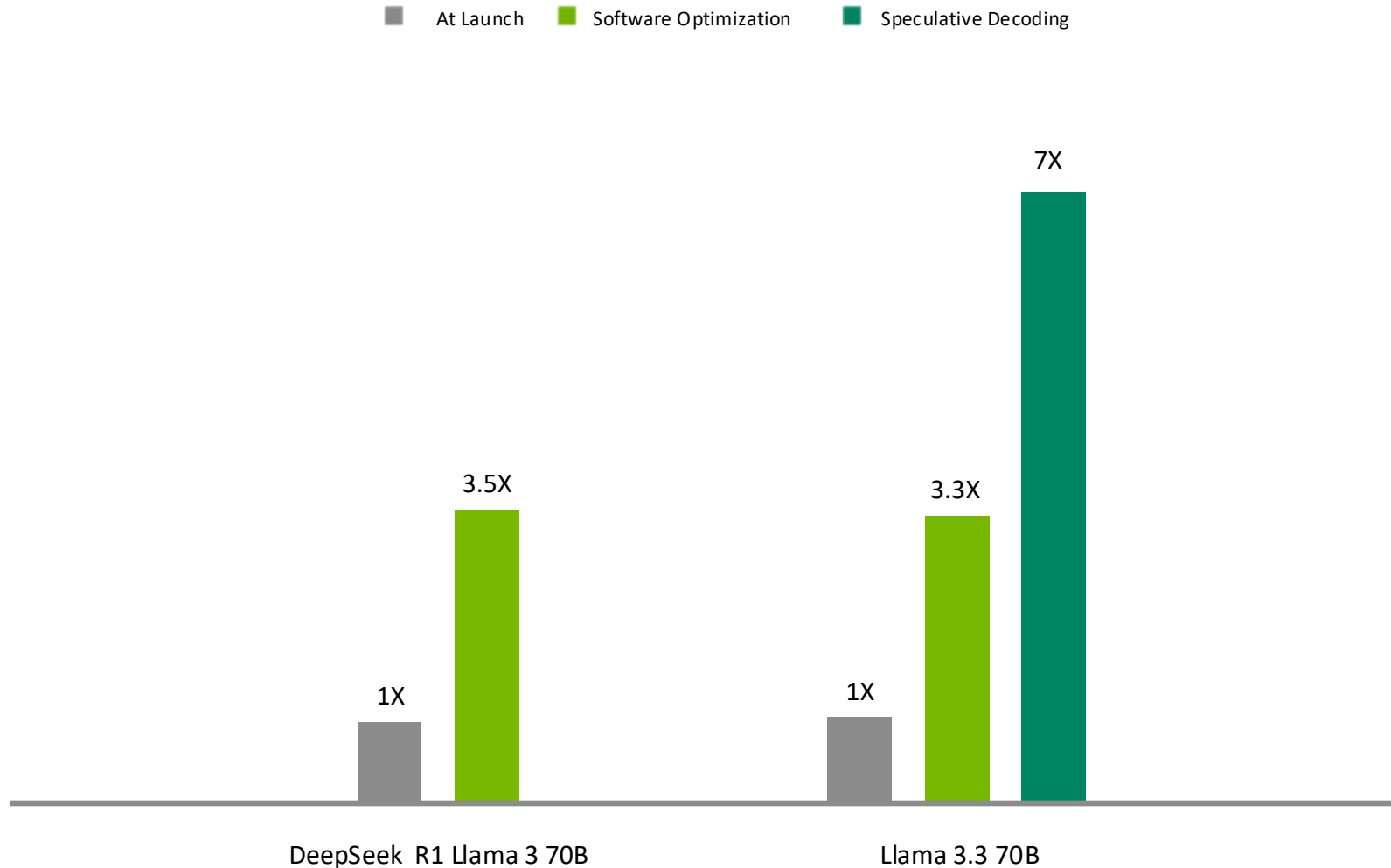
	Officially Supported	Community Supported
Inference Serving	PyTorch, VLLM, SGLang	MLC, Llama.cpp, Ollama, HuggingFace Transformers
Cadence of Release	PyTorch, VLLM, SGLang - Monthly	Varies (as needed with new versions)
Release Mechanism	NGC (Containers) Developer.nvidia.com (Wheels)	Jetson AI Lab (jetson-containers)
Support	Nvidia supported	Community supported
For Production	Recommended	For developers



`nvcr.io/nvidia/vllm:25.09-py3`

Accelerate Jetson Thor Performance Post-Launch

3.5X Increase with software optimization and 7X Increase with speculative decoding



Benchmark Tutorial: https://www.jetson-ai-lab.com/tutorial_gen-ai-benchmarking.html

Continuously Accelerating Gen AI on Jetson Thor

New Model Support

- Day 0 support for many models.
- Gpt-oss :
 - Gpt-oss supported on Llama.cpp on Day 0.
 - VLLM container released with support of gpt-oss models within few weeks.
- Nemotron:
 - All Nemotron Nano Models supported on Thor with all supported quantization's

New NVFP4 Support

- Support for all quantization's.
- NVFP4 support:
 - Many nvfp4 checkpoints supported in vLLM for models like Llama and Qwen.
- Performance of these models will keep improving in future as new nvfp4 kernels are enabled and optimized.

Speculative Decoding

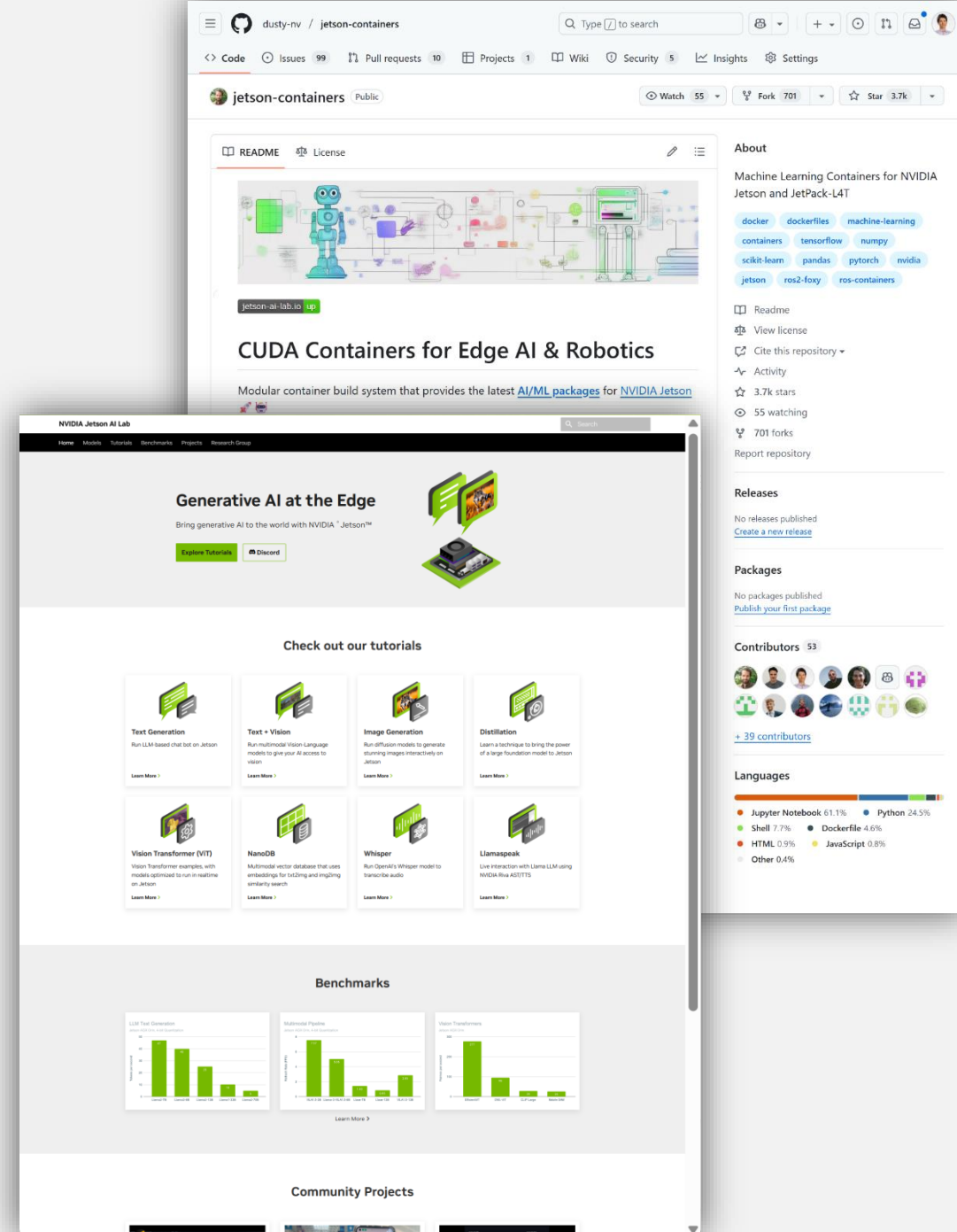
- Support for Eagle 3 Speculative Decoding in VLLM container.
- Up to 7X speed up by enabling speculative decoding and software optimization post-launch.
- Example:
 - Llama 3.3 70B W4A16 with Speculative Decoding

Model	At Launch (Aug 2025)	SD+ SW Optimization (Sep 2025)
Llama 3.3 70B (W4A16)	12.6 Tokens/sec	88.62 Tokens/sec

Jetson AI Lab

Integral part of Jetson developer experience

- Jetson AI Lab website
 - NVIDIA is actively working on to add Thor support
 - Thor benchmark tutorial is up
 - 1 of Top 10 shows Thor instruction, 2 other confirmed to work on Thor
- “jetson-containers” tool
 - NVIDIA is committed to support jetson-containers tool
 - *Migration work underway*
 - Jetson PyPI server already migrated
 - CI/CD pipelines with Orin, Thor runners
 - Container Registry migration planned in Nov.



Jetson Thor Roadmap



Released

In development

In planning

Products in development and planning are subject to change.

NVIDIA Isaac GROOT Hardware in the Loop

Robot Brain (GROOT N1)

Real-time inference on **Jetson Thor**



State
Commands

GR-1 Robot in Isaac Lab Simulation

Running on NVIDIA **RTX**



Demo-1

https://drive.google.com/file/d/1Xeuwb67RoMI1m-9u7m6s7GDakFYqe33v/view?usp=drive_link

Visual AI Agents Built with NVIDIA Metropolis VSS Blueprint

https://docs.nvidia.com/vss/latest/content/prereqs_thor.html



Demo-2

https://drive.google.com/file/d/1B-UK6NrWceuAQpP0f9G96U9IGRHh6FvK/view?usp=drive_link



Call to Actions

Explore [Jetson Thor webpage](#) and [launch technical blog](#).

Learn more about the product via [unboxing video](#).

Read Jetson Thor [Technical Brief](#) and New blog on faster inference is coming!

Available globally you can now get started with [Jetson AGX Thor Developer kit](#)!



Q & A